INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Annalakshmi R

Role: Data Preprocessing and Exploratory Data Analysis (EDA)

Project Title: Flight Delay Analysis and Visualization

WEEK 2 – Dataset Exploration and Understanding

During the second week, I focused on exploring the selected flight delay dataset to understand its structure and identify the variables that would be useful for analysis. I examined the dataset's columns, data types, and general organization to understand the information available for studying flight delays.

The dataset contained several flight-related attributes, including departure and arrival information, airline details, scheduled timings, and delay-related variables. I paid particular attention to numerical columns because they would be useful for statistical analysis and identifying patterns.

I also examined the distinction between different time-related variables, such as scheduled departure time, actual departure time,

scheduled arrival time, and actual arrival time. Understanding these fields was important because they could help explain how delays occurred and how they affected flight schedules.

Another part of my work involved identifying variables that might require preprocessing. I understood that some columns could contain missing values, inconsistent formats, or unusual numerical entries that needed to be investigated before analysis.

I coordinated with Ashvitha, who was responsible for data collection and data quality, to understand the initial observations regarding the dataset.

Outcome and Learning:

By the end of this week, I had developed a better understanding of the dataset's structure and its important variables. I learned how examining the dataset beforehand helps determine the preprocessing steps required for further analysis.